12.8 Solving One-Step Equations in One Variable Using Addition and Subtraction - Part 2

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• How can you use models to solve equations?
• How do you know two equations are equivalent?</td></tr>
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<u>Additional Terms:</u>
• Inverse operations
• Opposites</td><td colspan="2">• (Optional) Counters
• (Optional) Algebra tiles</td><td>• N/A</td></tr>
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- Students may struggle with understanding equivalent equations such as $x - 4 = 3$ is equivalent to $-4 + y = 3$.
- Students may struggle with integer operations.

</td><td>

- Draw a circle around the equal sign to emphasize that an equation is made of two equivalent expressions.
- Use diagrams and models to illustrate equations and the process of solving them as much as needed.
- As students continue to gain an understanding of solving equations students may struggle with solving equations with negative numbers. This struggle may be because they struggle with operations with integers or it may be because they do not grasp the mathematical meaning of an inverse. Reviewing the additive inverse property may be helpful. Students may struggle when equations are written in different forms. For example, $-2 + y = 3$ in 12.8.4 Exploration question 8, students may subtract 2 from both sides.

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<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
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MLR 2: Collect and Display

While pairs are working, circulate, collect, and make a visual display of vocabulary, phrases, and representations students use as they solve each situation. Make connections between how similar ideas might be communicated and represented in different ways.

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MA.K12.MTR.4.1: Discussion and Reflection

Students will be working to use inverse operations to solve one-step equations. Clarifications within this standard emphasize analyzing mathematical thinking and explaining methods and procedures. Encourage students to reflect on the processes they use and to describe them using correct mathematical vocabulary.

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<tr><td>Differentiate Instruction</td><td colspan="2">Operations with integers may still require a variety of entry points for students at different moments in their mathematics journey. For example, in the 12.8.1 Warm-Up, a student may know that the operation is addition and that the inverse is subtraction and therefore be able to answer all essential questions. However, when calculating, they may not get 25. Consider the strengths in this student and provide scaffolding for that student to still find success in this lesson. Whether using a calculator, accepting explanations to answers, or utilizing On-Ramp video learning tools and practice, consider additional entry points for learners at all stages in their journey.</td></tr>
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12.8.1 Warm-Up: Visualizing Solutions

Component Overview: *Students create a visual model to represent an equation. Students will then use the model and the equation to solve for the missing addend.*

12.8.1 Warm-Up: Visualizing Solutions

In the two previous lessons, visual representations such as number lines, bar models, hanger diagrams, balance scales, and drawings were used to represent equations.

1. Use the visual representation of your choice to represent the equation $37 = n + 12$.

   Answers will vary.

n	12
37	

2. Find the value of n.

   $n = 25$

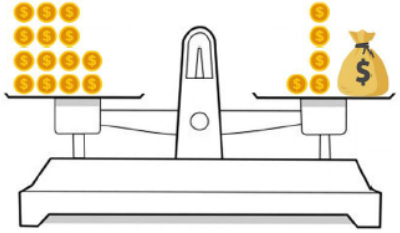
For students who have a good grasp of using inverse operations to solve one-step equations with addition or subtraction, challenge them to create a visual model for and solve a one-step equation using multiplication or division such as $6b = 42$ or $56 \div c = 7$.

12.8.2 Collaboration: Algebraic Equations From a Diagram

Component Overview: *Students collaborate to solve algebraic equations using the diagrams of a balance and a bar model. Students will draw on previous experiences with mathematical models and using inverse operations to solve one-step equations to complete this activity. Students may work with a partner to solve all the problems.*

12.8.2 Collaboration: Algebraic Equations from a Diagram

1. Ellen and Frey were given the balance scale shown below and told to write and solve an algebraic equation using inverse operations.



Ellen said, "There's no way to write an algebraic equation. If we can write an algebraic equation, we do not need inverse operations to solve it."

Frey said, "The only way to solve an algebraic equation is to use inverse operations to show the math is being reversed."

a. Whose statement do you agree with and why? Explain your answer to your partner.
Answers will vary. I agree with Frey's statement because inverse operations can be used to solve algebraic equations.

b. Write the algebraic equation represented by the balance scale.
$14 = 5 + x$

c. Solve the equation and show the steps you took.
$14 = 5 + x$
$14 - 5 = 5 + x - 5$
$9 = x$

d. How many gold coins are in the bag? Compare your answer with your partner.
9 gold coins

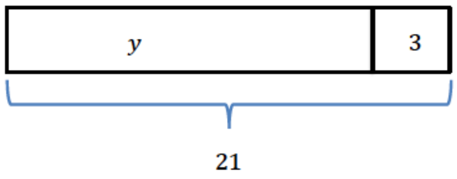
Use the vocabulary "algebraic equation" when discussing the activity, prompting students to give their ideas for what an algebraic equation can look like using references from what they have done in previous units.

If you notice students are struggling to make the connection between the balance scale and algebraic equations, ask questions to help guide their thinking:

- *How do we know that the amounts on the scales are equal?*
- *How is a balance scale like an algebraic equation?*
- *Do you know how much money is in the bag?*
- *How could you represent the bag of money in an algebraic equation?*

12.8.2 Collaboration: Algebraic Equations From a Diagram (continued)

2. A bar model is shown.



a. Write an addition equation and a subtraction equation that can be represented by the bar model.

Addition Equation	Subtraction Equation
$y + 3 = 21$	$21 - 3 = y$

b. Describe the relationship between the two equations.
Answers will vary. The relationship between the two equations is that inverse operations can be used to rewrite one equation so that it is written like the other equation.

c. Find the value of y using either equation.

$y = 18$

d. Check your solution with your partner. Make any revisions if necessary.

$18 + 3 = 21$
$21 = 21$

Differentiation is built in throughout this activity. The balance scale and bar models provide visual entry points, allowing students to access the content through diagrams. Discussion is built in through partner work, providing additional auditory entry points. Consider having students use counters to model one or both problems. Students will physically move the counters to demonstrate performing the same action to both sides of an equation. This will provide an additional kinesthetic entry point.

Ensure students are making the connection between the visual models and the algebraic equations to build conceptual understanding. If you notice students are still struggling with the equations, ask questions like:

- *How does the addition equation relate to the bar model?*
- *How does the subtraction equation relate to the bar model?*
- *How are the equations related to each other?*
- *Why is the value of the variable the same in both equations?*
- *How can you check your solutions?*

12.8.3 Guided Practice: Solving One-Step Algebraic Equations Using Inverse Operations

Component Overview: *Students use inverse operations of addition and subtraction to solve for the missing value in a one-step equation. This activity builds on students' previous experiences with inverse operations.*

12.8.3 Guided Practice: Solving One-Step Algebraic Equations Using Inverse Operations

1. Use the equation $31 = m - 19$ to answer the following.

   a. What operation is being performed in the equation?
   Subtraction

   b. What is the inverse operation?
   Addition

   c. Solve the equation using the inverse operation to find the value of m. Show your work.
   $31 = m - 19$
   $31 + 19 = m - 19 + 19$
   $50 = m$

   d. Verify that the solution is correct by substituting the value found for m into the original equation.
   $31 = 50 - 19$

2. Use the equation $d + 64 = {}^{-}12$ to answer the following:

   a. What is the inverse operation that will be used to solve for d?
   Subtraction

   b. Write and solve the equation using the inverse operation.
   $d + 64 - 64 = -12 - 64$
   $d = -76$

   c. What is the value of d?
   $d = -76$

   d. Check your solution with your partner.
   $-76 + 64 = -12$
   $-12 = -12$

Review key vocabulary terms such as "equation, operation, inverse, value," and "substitute" with students. Have students point out examples of each in a mathematical problem. Encourage students to explain the terms in their own words and use precise mathematical vocabulary.

Consider asking the following questions to further scaffold instruction throughout this section:

- *What do you notice about the equation?*
- *What is the inverse operation? How does using the inverse operation balance the equation?*

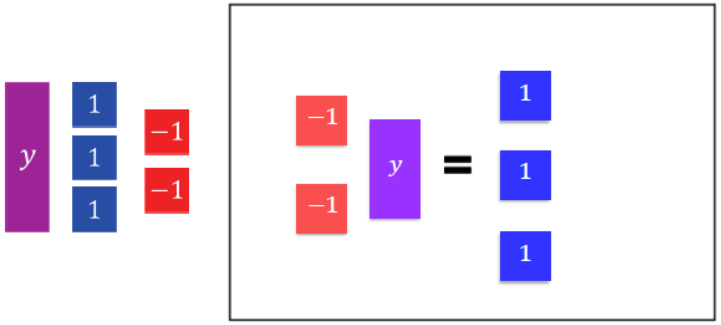
12.8.4 Exploration: Solving Equations Using Opposites

Component Overview: *Students use algebra tiles to model solving equations. Additionally, students explore solving an equation with a negative number. The understanding of opposites and inverses should develop to additive inverse. Encourage students to work with a partner and to discuss and explain each step of the process with each other.*

12.8.4 Exploration: Solving Equations Using Opposites

Work with your partner to complete the following.

1. Arrange the algebra tiles provided on the equation mat so that they represent the equation $-2 + y = 3$.



2. Check your diagram with your partner.
   Answers will vary.

3. For this equation, there are no tiles that can be removed from both sides of the equation to find the value of y.
   Discuss with your partner how zero pairs could be used to find the value of y.
   Answers will vary. Zero pairs can be used by adding 2 "positive one" representations to both sides of the equation.

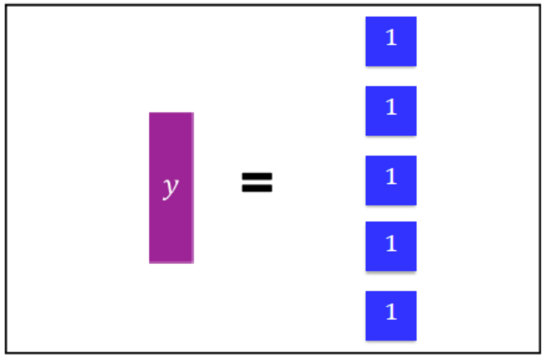
4. The goal when solving an equation is to isolate the variable. Which tiles can be added to both sides of the equation so that all tiles other than y on the left side of the equation can be eliminated?
   2 blue tiles

As students discuss zero pairs, they should be explaining the concept of additive inverse. Students do not need to name the property but by adding two positive tiles to solve the student is using the additive inverse. It may be helpful for some students to see this written. $^-2 + 2 = 0$. *If students choose to use subtraction of negative two, they may struggle finding* $3 - {}^-2$ *because they struggle with operations with integers. This is why using the concept of zero pairs is used here. Continue to help students grasp the concept by using the terminology of zero pairs and additive inverse.*

12.8.4 Exploration: Solving Equations Using Opposites (continued)

5. Explain why, when tiles are added to one side of the equation, they also must be added to the other side.
Answers will vary. The same number of tiles must be added to both sides because the value on each side of the equation must remain equivalent and adding or subtracting the same amount on each side will maintain equivalency.

6. Draw tiles on the equation mat to represent the solution to the equation.



7. What is the value of y?
$y = 5$

8. Discuss with your partner how solving for the variable in the equation $-2 + y = 3$, was different from solving for the variable in the equation $d + 64 = {}^{-}12$, even though both equations include addition. Summarize your discussion.
Answers will vary. Solving for the variable for the equation $d + 64 = {}^{-}12$ involved using inverse operations with just numbers and solving for the variable for the equation $-2 + y = 3$ involved using inverse operations with modeling and zero pairs.

Consider having students use algebra tiles to model the problem. Students will physically move the tiles to demonstrate performing the same action to both sides of the equation. Have students verbally explain the process they are using to solve the equation.

This can lead to a discussion about the fact that $-2 + y = 3$ and $y - 2 = 3$ are equivalent equations. When students are stuck on using inverse operations, they will often try to solve an equation like $-2 + y = 3$ by subtracting -2 from both sides of the equation because "it's the inverse operation." Students need to understand that inverse operations are helpful to isolate variables, but they can also consider the terms and use their knowledge of creating zero pairs to eliminate terms.

To provide additional scaffolding for this section, ask questions like:

- *What process do you use when solving an equation using zero pairs?*
- *What process do you use when solving an equation with inverse operations?*
- *How are these approaches similar? How are they different?*
- *Do you think one approach would be better to use in certain situations? When? Why?*

12.8.5 Collaboration: Solving Equations

Component Overview: *Students work with a partner to solve several one-step equations using addition or subtraction. Students can solve each equation with their partner, or each partner may solve one column of equations and then explain the process they used to their partner.*

12.8.5 Collaboration: Solving Equations

With your partner, determine who will be Partner A and who will be Partner B.

1. Solve the equations from your designated column.

Partner A	Partner B
$18 = 8 + x$ $18 - 8 = 8 - 8 + x$ $x = 10$	$-8 = -18 + x$ $-8 + 18 = -18 + 18 + x$ $x = 10$
$p - 9 = 16$ $p - 9 + 9 = 16 + 9$ $p = 25$	$9 = p - 16$ $9 + 16 = p - 16 + 16$ $p = 25$
$469 = 456 + w$ $469 - 456 = 456 - 456 + w$ $w = 13$	$-469 + w = -456$ $-469 + 469 + w = -456 + 469$ $w = 13$

2. The equations in each row have the same solutions. Check your work with your partner, making revisions to your work if necessary.

$18 = 8 + 10$	$-8 = -18 + 10$
$25 - 9 = 16$	$9 = 25 - 16$
$469 = 456 + 13$	$-469 + 13 = -456$

As you circulate listen for students using terminology such as zero pairs or additive inverse for the equations with negative numbers on the side with the variable.

For students who finish early ask them to write an explanation as to why each of the questions are the same answer. Look for students who understand that this distance between the values in the equations are the same.

Encourage students to explain the strategies they used to solve the equations. To support students, provide sentence frames such as, "I noticed that ______, so I ______." Or: "First, I ______ because ______."
When students share their answers with a partner, prompt them to rehearse what they will say when they share with the full group. Rehearsing provides opportunities for students to clarify their thinking.

12.8.6 Your Turn!

Component Overview: *Students practice solving one-step equations using addition or subtraction. Students may use inverse operations or opposites to solve the equations. Allow students to use visual models as needed.*

12.8.6 Your Turn!

1. Solve each equation. Show your steps in finding the solution.

   a. $h + 37 = 62$

   $h + 37 - 37 = 62 - 37$

   $h = 25$

   b. $18 = y - 23$

   $18 + 23 = y - 23 + 23$

   $41 = y$

   $y = 41$

If students are struggling to solve the equations, ask questions like the following to provide further scaffolding:

- *What strategy can you use to solve the equation?*
- *How can you isolate the variable?*
- *What operation did you do to both sides of the equation?*
- *Why do you perform the same action on both sides of an equation?*

12.8.7 Wrap-Up: Cool Down

Component Overview: *Consider using the Wrap-Up as a formative assessment to gather data on student progress and vocabulary understanding from this lesson. This Wrap-Up assesses students' abilities to explain how to use inverse operations to solve a one-step equation using addition or subtraction.*

12.8.7 Wrap-Up: Cool Down

1. Your friend Giselle is absent today and will need the notes from class. Write a quick note to help her understand how to use inverse operations to solve the equation $25 = y + 7$.

   Answers will vary. To solve the equation using inverse operations, use the inverse of addition. Subtract 7 from both sides of the equation. Then check your work by substituting your solution into the original equation to check for equivalent values on both sides of the equal sign.

Students may also use terminology such as zero pairs or additive inverse.

For students who might struggle to represent their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.